

Gaussian random vectors

- Multivariate normal distributions

Multivariate normal

A multivariate normal random variable is denoted with

$$w \sim \text{Normal}_D(\mu, V)$$

This variable attains real vectorial values in $\mathbb{R}^D$

The probability density function is given by

$$p(w) = \text{Normal}_D(w; \mu, V) = \frac{1}{\sqrt{(2\pi)^D |V|}} \exp\left(-\frac{(w - \mu)^t V^{-1} (w - \mu)}{2}\right)$$

The parameters are $\mu \in \mathbb{R}^D$ and $V \in \mathbb{R}_{sym,+}^{D \times D}$... $\mathbb{R}_{sym,+}^{D \times D}$ denotes symmetric, strictly positive definite matrices

By convention:

- vectors in $\mathbb{R}^D$ are column arrays
- the transpose v^t is a row array
- V^{-1} denotes the inverse matrix
- $(w - \mu)^t V^{-1} (w - \mu)$ is an inner product on $\mathbb{R}^D$
- $|V|$ denotes the determinant on $\mathbb{R}^{D \times D}$

...this is a linear map $\mathbb{R}^D \mapsto \mathbb{R}$

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...for scalars c , the determinant follows $|cV| = c^D |V|$

Special cases

We consider a variable

$$w \sim \text{Normal}_D(\mu, V),$$

$$\mu \in \mathbb{R}^D,$$

$$V \in \mathbb{R}_{sym,+}^{D \times D}$$

- In **1D**, this is a univariate normal and reduces to the usual normal

$$\text{Normal}_1(\mu, V) \equiv \text{Normal}(\mu, V)$$

$$\mu \in \mathbb{R}^1 \equiv \mathbb{R},$$

$$V \in \mathbb{R}_{sym,+}^{1 \times 1} \equiv (0, \infty)$$

- In **2D**, this is a bivariate normal

$$\text{Normal}_2(\mu, V)$$

$$\mu \in \mathbb{R}^2,$$

$$V \in \mathbb{R}_{sym,+}^{2 \times 2}$$

This is often parametrized by

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad V = \begin{bmatrix} v_1 & \rho\sqrt{v_1v_2} \\ \rho\sqrt{v_1v_2} & v_2 \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{v_1} & 0 \\ 0 & \sqrt{v_2} \end{bmatrix}}_{\text{diagonal with scales}} \underbrace{\begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}}_{\text{square with correlations}} \underbrace{\begin{bmatrix} \sqrt{v_1} & 0 \\ 0 & \sqrt{v_2} \end{bmatrix}}_{\text{diagonal with scales}}$$

for $\mu_1, \mu_2 \in \mathbb{R}$ and $v_1, v_2 \in (0, \infty)$ and $\rho \in (-1, +1)$

Simulation

To generate a multivariate normal variate

$$w \sim \text{Normal}_D(\mu, V)$$

- First, **find** a square-root L such that

$$LL^t = V$$

- Then, **generate** standard normal variates

$$\zeta_d \sim \text{Normal}(0, 1), \quad d = 1, \dots, D$$

- Finally, **transform**

$$w = \mu + L\zeta, \quad \zeta = \begin{pmatrix} \zeta_1 \\ \vdots \\ \zeta_D \end{pmatrix}$$

The algorithm works because of the **de-whitening property**

$$\left. \begin{array}{l} \zeta \sim \text{Normal}_K(0, I) \\ w = \mu + L\zeta \end{array} \right\} \implies w \sim \text{Normal}(\mu, LL^t)$$

for any $L \in \mathbb{R}^{D \times K}$

In this algorithm, we have **many choices**

- dimension K ...clear-cut decision $K \geq D$
- square-root L ...not a clear-cut decision

Transformation properties

De-whitening property

$$\left. \begin{array}{l} \zeta \sim \text{Normal}_K(0, I) \\ w = \mu + L\zeta \end{array} \right\} \implies w \sim \text{Normal}_D(\mu, LL^t)$$

Whitening property

$$\left. \begin{array}{l} x \sim \text{Normal}_D(\mu, LL^t) \\ y = L^{-1}(x - \mu) \end{array} \right\} \implies y \sim \text{Normal}_D(0, I)$$

General transformation property

$$\left. \begin{array}{l} x \sim \text{Normal}_K(m, V) \\ y = a + Bx \end{array} \right\} \implies y \sim \text{Normal}_D(a + Bm, BVB^t)$$

Here, the dimensions are

$$m \in \mathbb{R}^K$$

$$V \in \mathbb{R}^{K \times K}$$

$$a \in \mathbb{R}^{D \times 1}$$

$$B \in \mathbb{R}^{D \times K}$$

Other properties

Addition property

$$\left. \begin{array}{l} x_1 \sim \text{Normal}_D(m_1, V_1) \\ x_2 \sim \text{Normal}_D(m_2, V_2) \\ y = x_1 + x_2 \end{array} \right\} \implies y \sim \text{Normal}_D(m_1 + m_2, V_1 + V_2)$$

Magnitude properties

$$\left. \begin{array}{l} x \sim \text{Normal}_D(0, I) \\ y = \|x\|_2^2 \end{array} \right\} \implies y \sim \chi_D^2$$

$$\left. \begin{array}{l} x \sim \text{Normal}_D(0, I) \\ z = \|x\|_2 \end{array} \right\} \implies z \sim \chi_D$$

here $\|\cdot\|_2$ is the Euclidean norm on $\mathbb{R}^D$

Covariance factorizations

A covariance matrix $V \in \mathbb{R}_{sym,+}^{D \times D}$ attains three **important decompositions**:

- **Lower Cholesky decomposition**
Our matrix is factorized like this

...in Matlab this is `L = chol(V, 'lower');`
...runs in $\approx \frac{1}{3}D^3$ flops

$$V = LL^t$$

where $L \in \mathbb{R}^{D \times D}$ is lower triangular

- **Eigen-decomposition**
Our matrix is factorized like this

...in Matlab this is `[U,S] = eig(V);`
...runs in $\approx 9D^3$ flops

$$V = USU^t$$

where $S \in \mathbb{R}^{D \times D}$ is diagonal and $U \in \mathbb{R}^{D \times D}$ is orthogonal

- **Singular-value decomposition**
Our matrix is factorized like this

...in Matlab this is `[W,S,~] = svd(V);`
...runs in $\approx 22D^3$ flops

$$V = WSW^t$$

where $S \in \mathbb{R}^{D \times D}$ is diagonal and $W \in \mathbb{R}^{D \times D}$ is orthogonal

Covariance square-roots

We consider a covariance matrix $V \in \mathbb{R}_{sym,+}^{D \times D}$ and its three **important decompositions**:

Lower Cholesky decomposition

$$V = LL^t$$

we have our square-root **directly**

This method:

- is **little** expensive
- usually fails

Eigen-decomposition

$$V = USU^t$$

we **derive** our square-root by

$$L = U\sqrt{S}$$

This method:

- is moderately expensive
- often fails

Singular-value decomposition

$$V = WSW^t$$

we **derive** our square-root by

$$L = W\sqrt{S}$$

This method:

- is very expensive
- **never** fails

Bayesian considerations

In BO, we will consider multivariate normals in this form

$$\begin{aligned}m &\sim \text{Normal}_D (M, C) \\ w|m &\sim \text{Normal}_K (a + Bm, V)\end{aligned}$$

This is a Bayesian model which is **conjugate**

The posterior is

$$m|w \sim \text{Normal}_D (M', C')$$

and the **hyper-parameters** are

$$\begin{aligned}C' &= \left(C^{-1} + B^t V^{-1} B\right)^{-1} \\ M' &= C' \left(C^{-1} M + B^t V^{-1} (w - a)\right)\end{aligned}$$